

WEEK 8 – STUDENT PERFORMANCE DASHBOARD

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INTRODUCTION:

In this project, Power BI Desktop is used to analyze student performance based on examination scores and other student-related factors.

The dashboard helps analyze average scores, subject-wise performance, grades, and pass percentages. Interactive slicers and a drill-through page are also used to explore student performance in detail. The dataset prepared during Week 7 is used to create the visualizations.

OBJECTIVES:

The main objectives of this project are:

- To analyze the overall academic performance of students.
- To calculate and display average examination scores.
- To compare Mathematics, Reading, and Writing scores.
- To analyze performance based on parental education.
- To compare students who completed test preparation with those who did not.
- To analyze academic performance based on lunch type.
- To display the distribution of students across different grades.
- To create interactive slicers for filtering the dashboard.

DATA ANALYSIS:

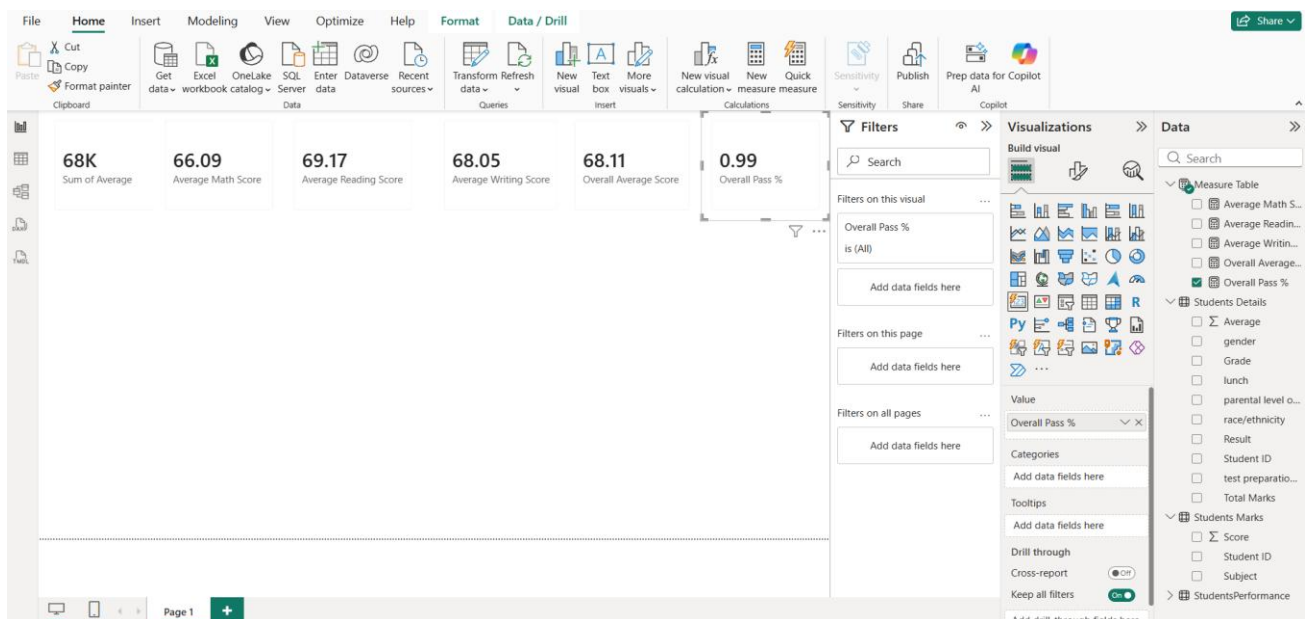
The prepared Student Performance in Exams dataset from Week 7 is used for analysis. The dataset contains student information, subject scores, parental education,

lunch type, test preparation status, average scores, grades, and pass/fail classifications.

Power BI visualizations and existing DAX measures are used to compare academic performance and present the results clearly. The dashboard allows users to explore different student groups and identify patterns in the dataset.

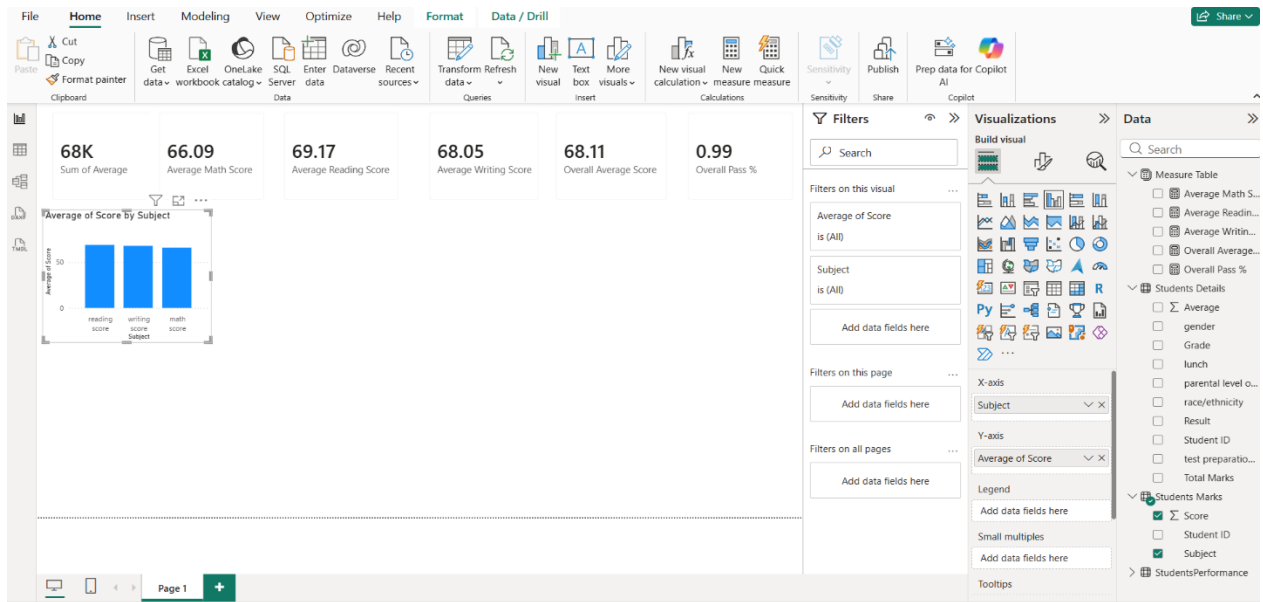
KPI CARDS:

Five Card visuals were created to display the important performance measures. First, a Card visual was selected from the Visualizations pane. The Average Score measure was dragged into the card. The same procedure was followed to create cards for Average Math Score, Average Reading Score, Average Writing Score, and Overall Pass %. Titles were added to identify each card clearly. These cards provide a quick summary of student performance.



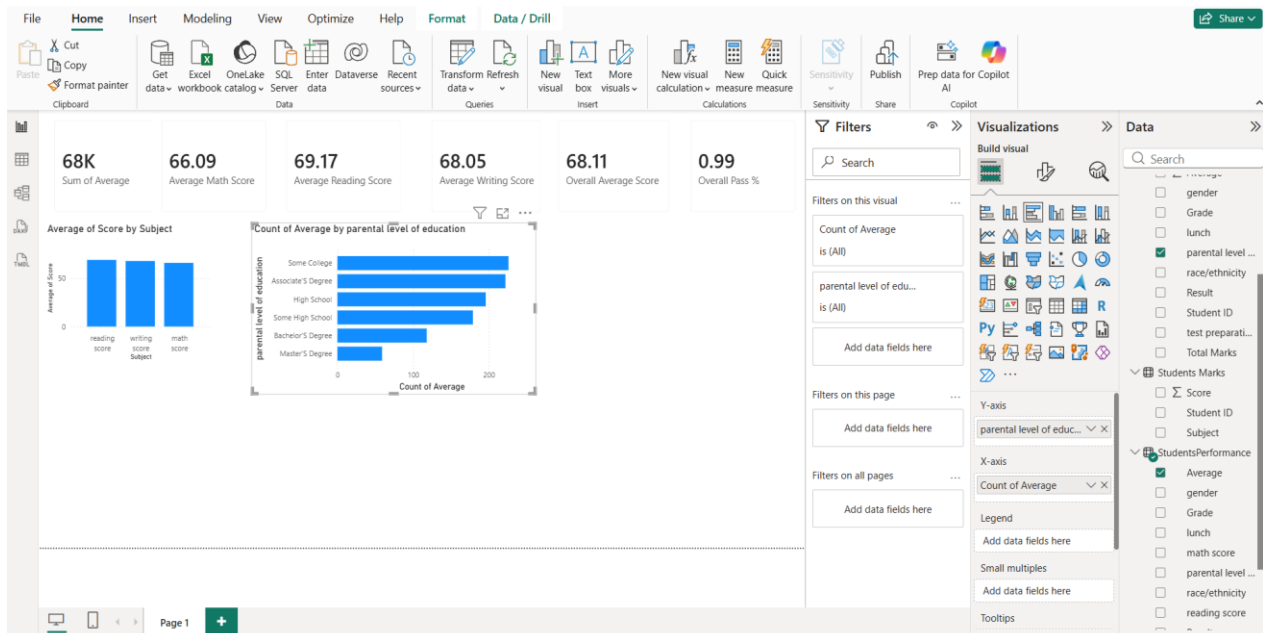
SUBJECT PERFORMANCE VISUALIZATION:

A Clustered Column Chart was selected from the Visualizations pane to compare the average scores of different subjects. The Subject field was dragged into the X-axis, and the Score field was placed in the Y-axis. The aggregation of Score was set to Average. The chart title was changed to Average Score by Subject. This visualization helps compare Mathematics, Reading, and Writing performance and identify the subject with the highest or lowest average score.



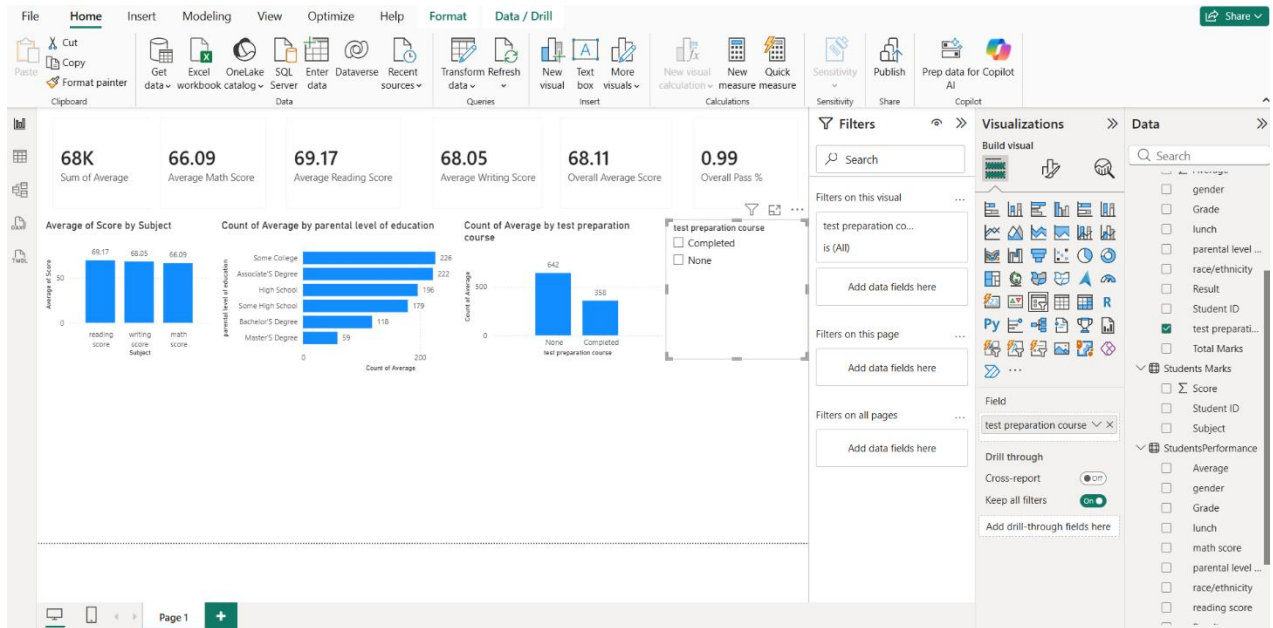
PARENTAL EDUCATION ANALYSIS:

A Clustered Bar Chart was selected to analyze student performance based on parental education. The Parental Education Level field was dragged into the Y-axis, and the Average Score measure was placed in the X-axis. The chart title was changed to Average Score by Parental Education. The visual allows comparison of average student scores across different parental education categories.



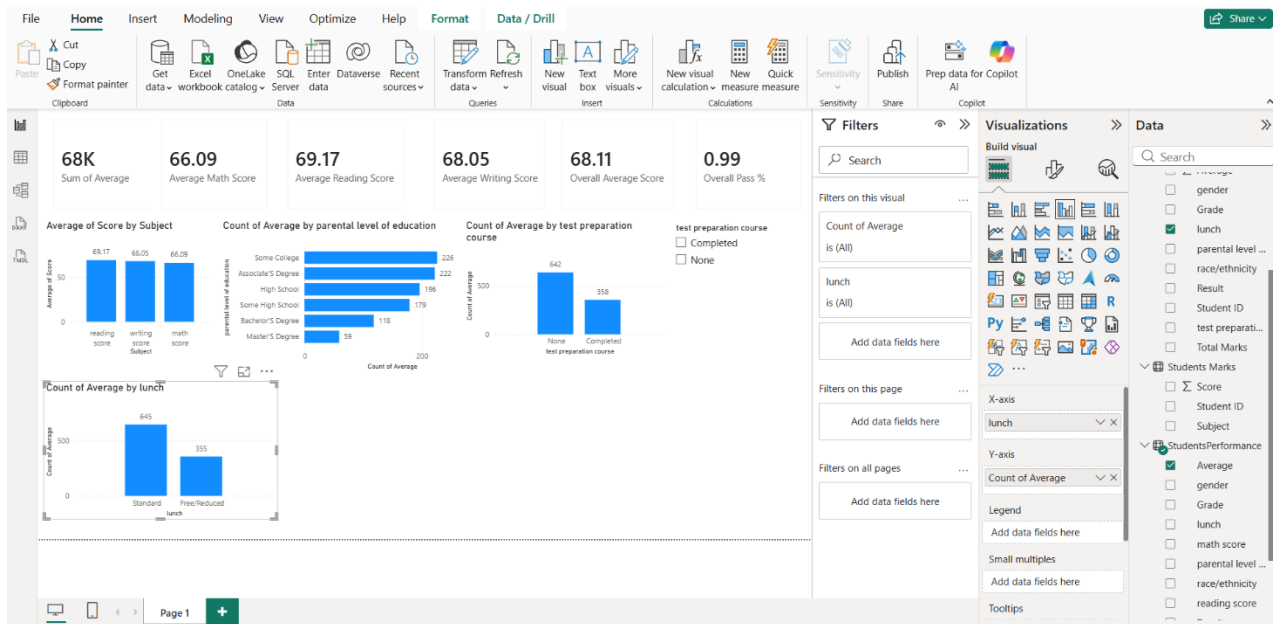
TEST PREPARATION ANALYSIS:

A Clustered Column Chart was created to compare students who completed the test preparation course with those who did not. The Test Preparation Course field was placed on the X-axis, and the Average Score measure was placed on the Y-axis. The chart title was set to Average Score by Test Preparation. A separate slicer was also created using the Test Preparation Course field.



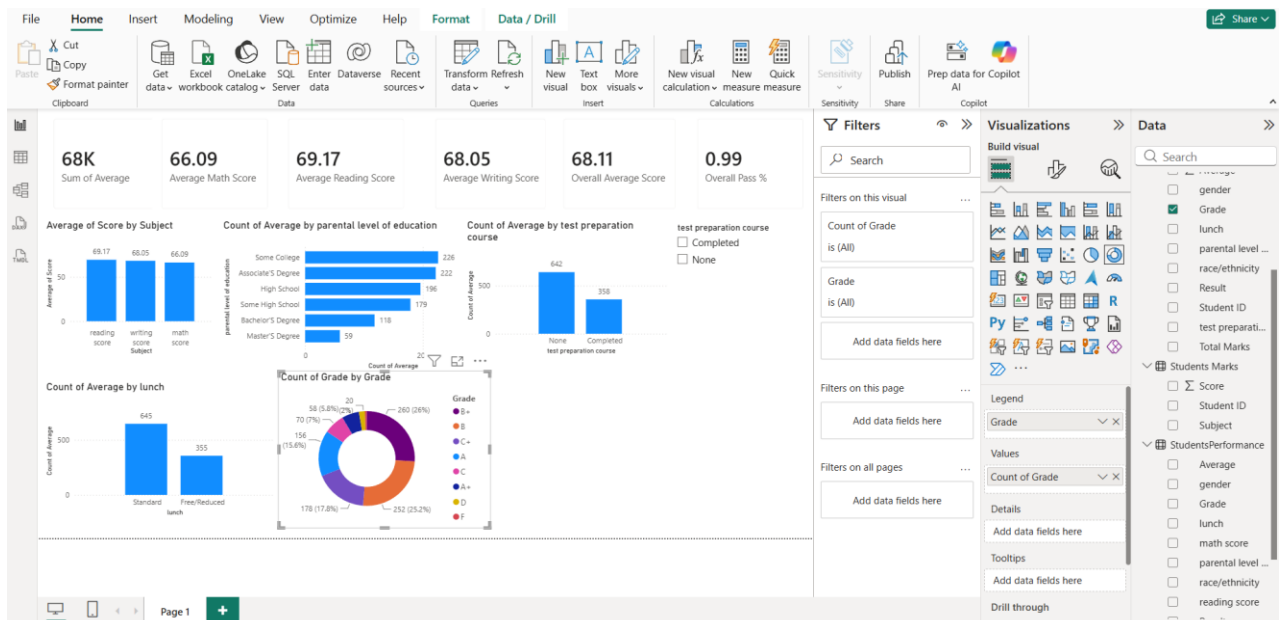
LUNCH TYPE ANALYSIS:

A Clustered Column Chart was selected to compare average scores based on lunch type. The Lunch field was dragged into the X-axis, and the Average Score measure was placed in the Y-axis. The title was changed to Average Score by Lunch Type. The visualization compares the average scores of students in the available lunch categories.



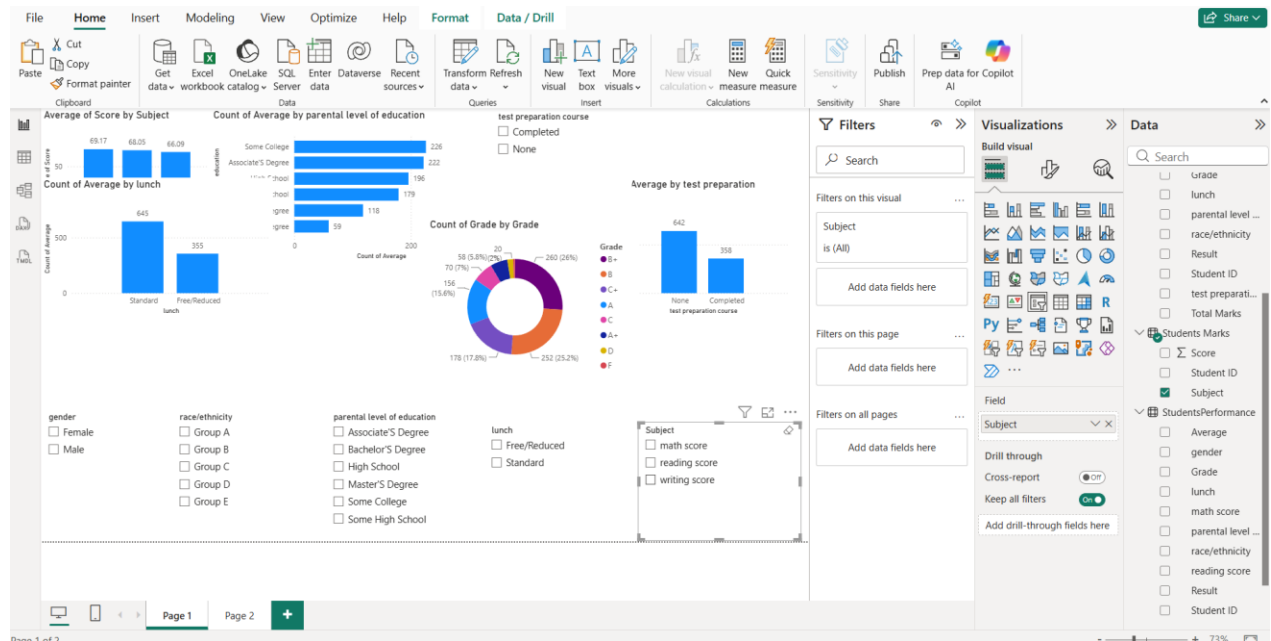
GRADE DISTRIBUTION:

A Donut Chart was created to display the distribution of student grades. The Grade field was placed in the Legend section, and the Grade field was added to Values with the Count aggregation. The chart title was changed to Grade Distribution. The donut chart displays the number of students in each available grade category and helps understand the overall distribution of student performance.



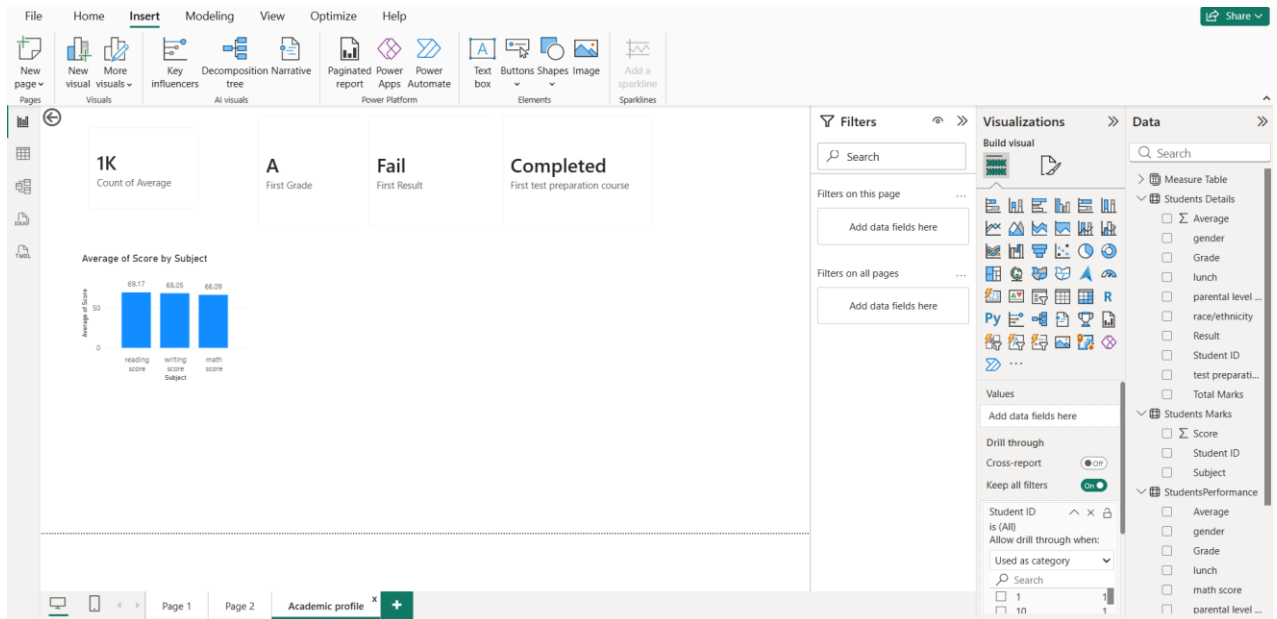
INTERACTIVE SLICERS:

Slicer visuals were added to make the dashboard interactive. The Gender, Race/Ethnicity, Parental Education, Lunch, Test Preparation Course, and Subject fields were added to separate slicers. When a category is selected, the related charts and cards are filtered according to the selection. This helps users examine student performance for different groups.



ACADEMIC PROFILE DRILL-THROUGH:

A new report page named Academic Profile was created for student-level analysis. The available student identifier field was added to the Drill-through section. An Average Score card, a subject performance chart, a Grade card, a Result card, and a Test Preparation card were added to the page. A Back button was inserted to return to the main dashboard. This page allows users to view the academic details of an individual student when the required student-level field is available.



CONCLUSION:

This project helped develop practical skills in creating Power BI visualizations and interactive dashboards. The different charts and KPI cards provide a clear way to examine student performance, while slicers and drill-through functionality allow users to explore the data in more detail.